

Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering

09/2023 – Present

Kabataş Boys' High School

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00.
- Actively participated in various extracurricular activities and student societies, including Debate, Model United Nations, Entrepreneurship, and the Coding Club.

Experience

Executive Board Member at Radyo Boğaziçi

09/2023 – Present

- Developed strong skills in team coordination, social media management, budget planning, event organization, communication, and operational decision-making.

Work and Travel Program Participant

06/2025 – 09/2025

- Gained valuable international experience through the cultural exchange and work program in the USA which improved my communication, English, teamwork and problem-solving skills.

Regional Organizational Team Member at ESTIEM

09/2023 – Present

- Contributed to the planning and coordination of events, team communication, and organizational processes within ESTIEM, an international network connecting Industrial Engineering and Management students across Europe.

Technical Projects

AI-Powered FinTech Intelligence Platform — Next.js, FastAPI, PostgreSQL, Docker

- Architected a full-stack, deployment-ready financial dashboard combining personal finance analytics, an algorithmic portfolio risk evaluator, and OpenAI-generated insights with secure NLP fallbacks.
- Engineered a live limit-order trading simulator featuring a synchronous Price-Time Priority matching engine and achieved 100% backend test coverage with automated CI/CD pipelines.

SchedSim CPU Scheduler — x86-64 Assembly, GNU Toolchain, Linux Syscalls

- High-performance CPU scheduling simulation engine (FCFS, SJF, SRTF, PF, RR) built with zero standard library dependencies, utilizing direct Linux syscalls for I/O and process management.

Spirelayer Command Interpreter — C, Custom Backtracking Lexer

- Robust CLI-based interpreter replicating fantasy RPG REPL mechanics; implemented an autonomous backtracking lexical parser to handle non-deterministic grammar and complex command resolution.

Aura Functional Interpreter — Python, Functional Paradigms, Dual Scoping Engine

- Advanced interpreter for a custom functional language supporting first-class closures, higher-order function patterns, and toggleable lexical/dynamic scoping engines for complex variable resolution.

MatrixNet — Java, Tarjan's SCC Algorithm, Graph Theory

- Network graph analysis tool implementing Tarjan's SCC and Articulation Point detection ($O(V + E)$); optimized with Open Addressing to eliminate internal JVM string allocation overheads.

GigMatch-Pro — Java, Memory-Optimized Heaps

- Scalable task-matching marketplace engine utilizing optimized dynamic Max-Heaps for high-efficiency priority allocation and candidate-to-job matching algorithms.

Nightpass Survival Game — Java, Rule Engine Architecture

- Algorithmic implementation of a complex rule engine governing strategic deck, hand, and resource economics in a terminal-based survival card game environment.

Volunteer Experience

Education Mentor at TOÇEV

09/2019 – 03/2020

- Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds.

Volunteer at Kabataş Boys’ High School Community Involvement Projects Club

09/2018 – 06/2023

- Carried out various volunteer projects aimed at helping and supporting people in need through community involvement initiatives.

Volunteering Division Member at Boğaziçi University Industrial Engineering Club

09/2023 – 06/2024

- Supported the organization and execution of volunteer initiatives and community-oriented projects within the club.

Education Mentor at Derece Kampüsü

- Mentored high school students for the university entrance exams, providing academic guidance and exam strategy support.

Technical Skills

- **Low-Level & Systems:** x86-64 Assembly, C, Linux Syscalls, Process Scheduling, Multithreading (Pthreads), Memory Management
- **Enterprise Full-Stack:** Next.js 15+, TypeScript, React, Prisma (ORM), FastAPI, Tailwind CSS, Framer Motion, PostgreSQL, SQLite
- **Data Structures & Algos:** Tarjan’s SCC, Articulation Points, Max-Heaps, Complexity Analysis (Big O), Graph Theory, Recursive Optimization
- **Paradigms & Models:** Functional (Scheme/Racket), Logic (Prolog), OOP (Java/C++), Interpreter Design, State Machines
- **AI & Modern Engineering:** Advanced Agentic Coding, LLM Prompting, AI-Assisted Architecture, Autonomous Feature Deployment
- **Tools & Ecosystem:** Linux/Unix, Git/GitHub, Docker, Bash, Alembic, CI/CD, Makefile, LaTeX, Adobe Photoshop

Languages

- Turkish (native)
- English (fluent)
- German (elementary)

Extracurricular Activities

- DJ, Radyo Boğaziçi
- Student-Athlete, Boğaziçi University Sailing Team
- Student-Athlete, Boğaziçi University Ski Team
- Windsurf Athlete, Competing Individually